

Strategy 3: CAM R3/S3 (89% WR)

Rules:

- 1 5-day trend DOWN or UP
- 2 ConsecDown/Up 0-2 (fresh trend, not exhausted)
- 3 Yesterday range > 2% and body ratio > 0.2 (not choppy)
- 4 Price touches R3 (for SHORT) or S3 (for LONG) within ATR tolerance
- 5 Bar closes below R3 / above S3 (bounce confirmed)
- 6 Enter at bar 10 close (10:15 AM)
- 7 Target: 1.75% | Stop: 1.50%
- 8 Trail: at +1.00% MFE, lock +0.075% profit
- 9 Runner: at target hit, don't close, trail with 0.25% step

Camarilla formulas:

- $R3 = \text{prev_close} + (\text{prev_high} - \text{prev_low}) \times 1.1 / 4$
- $S3 = \text{prev_close} - (\text{prev_high} - \text{prev_low}) \times 1.1 / 4$

Why it works: Camarilla levels are where institutional orders cluster. Fresh downtrend + strong yesterday + bounce at R3 = high probability the stock continues falling.

Backtest: 1,011 trades, 89% WR. Walk-forward: Train 88.7%, Test 89.8%.

Strategy 4: PIVOT R1/S1 (96% WR)

What: Same as CAM but using Classic Pivot R1/S1 levels instead. Only fires when CAM doesn't (fallback).

Rules:

- Same filters as CAM (trend, CD 0-2, yesterday)
- CAM checked FIRST. Pivot only if no CAM signal for that stock.
- Price touches Pivot R1 (for SHORT) or S1 (for LONG)
- Target: 0.75% | Stop: 1.50%

Pivot formulas:

- $PP = (\text{prev_high} + \text{prev_low} + \text{prev_close}) / 3$
- $R1 = 2 \times PP - \text{prev_low}$
- $S1 = 2 \times PP - \text{prev_high}$

Why it works: R1 is ABOVE R3 (different price zone). Some stocks overshoot R3 and reach R1. These are extra signals CAM misses.

Backtest: 282 trades total, 81 new (CAM missed), 96% WR. Walk-forward: Train 96.2%, Test 94.6%.

Daily Flow

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9:15 AM · Bot starts (cron)
          Fetches capital from Upstox API
          Loads 10 days of 1-min data for 45 stocks
          Aggregates to 5-min bars
          Computes: trends, consec days, yesterday stats, MAs,
          volumes
9:20 AM · STRATEGY 1: GAP FILL scan
          Check 45 stocks for gaps > 2% + first bar reversal > 0.5%
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Enter MARKET orders for qualifying stocks
Monitor every 30 seconds (fast 0.5% target)
9:45 AM · STRATEGY 2: MA CONVERGENCE scan
Check 45 stocks for SMA20~EMA20 convergence + volume
condition
Skip stocks already in position from GAP
Enter MARKET orders for qualifying stocks
10:15 AM · STRATEGY 3: CAM R3/S3 scan
Check 45 stocks for R3/S3 touch + filters
Skip stocks already in position from GAP or MA
10:15 AM · STRATEGY 4: PIVOT R1/S1 scan (fallback)
Only for stocks where CAM didn't fire
Check for R1/S1 touch + same filters
10:16 AM to 3:00 PM · MONITOR all positions
Check every 60 seconds
Phase 1 (MFE < 1.0%): normal stop
Phase 2 (MFE >= 1.0%): trail activated, lock 0.075%
Phase 3 (target hit): runner mode, trail 0.25% step
3:00 PM · CLOSE ALL remaining positions
MARKET exit orders
3:01 PM · Write journal
Bot exits

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Position Management

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Entry:    MARKET order (fills instantly at best price)
Stop:     SL-M order (trigger price on exchange)
Exit:     MARKET order (price=0)
No duplicate positions:
- 1 stock = 1 trade maximum
- If GAP opens RELIANCE, no other strategy touches RELIANCE
Sizing:
- 20% of available capital per trade
- 5x broker leverage for equity
- Capital auto-fetched from Upstox at startup
Exit phases (CAM/Pivot only):
Phase 1 · MFE < 1.0%:
  Target 1.75%: close at target · runner mode
  Stop 1.50%: close at stop (loss)
  EOD: close at 3 PM
Phase 2 · MFE >= 1.0% (trail activated):
  Lock floor at +0.075% (breakeven + charges)
  If price reverses to 0.075%: exit (small win)
  If price continues to target: runner mode
Phase 3 · Target hit (runner mode):
  Don't close. Lock at target price.
  Trail with 0.25% step as price keeps moving.
  Exit when trail stop hit (at least +1.75%)
Exit (GAP/MA/Pivot):
  Fixed target and stop. No trail/runner.
  Close at target, stop, or EOD.

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Order Types

Action	Order Type	Price	Trigger	----- ----- ----- -----	Entry	MARKET	0	-		Stop loss	SL-M	0		stop price		Exit (target/trail/EOD)	MARKET	0	-	
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Backtest Results Summary

Strategy	Trades	WR	/trade (Rs 10L)	Walk-forward	----- ----- ----- -----	GAP FILL	643	99.8%	Rs 4,000	Train 99.7%	Test 100%		MA CONVERGENCE	75	91%	Rs 3,399	Train 90%	Test 94%		CAM R3/S3	1,011	89%	Rs 8,077	Train 88.7%	Test 89.8%		PIVOT R1/S1	81 (new)	96%	Rs 5,819	Train 96.2%	Test 94.6%	
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Server Details

Item	Value	----- -----	Server	GCP 136.111.68.229		SSH	ssh -i ~/.ssh/id_ed25519_gcp ai18developer@136.111.68.229		Cron	9:15 AM IST Mon-Fri		Kill switch	~/trading-bot/ENABLED		Token page	http://136.111.68.229		Bot version	v5 (4 strategies)	
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Daily Checklist

- 1 Before 9:15 AM: Refresh Upstox token (<http://136.111.68.229>)
 - 2 Bot runs automatically
 - 3 After 3 PM: Check results (./server_control.sh log)
 - 4 That's it
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Future Upgrade (at Rs 5L+ capital)

Option SELLING using CAM signals:

- 99% WR, Rs 1,823/trade
 - Sell CALL when CAM says SHORT
 - Sell PUT when CAM says LONG
 - Needs Rs 1.5L+ margin per lot
 - Save for when capital grows
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Generated: 2026-05-27 System validated and deployed on GCP server